

FraudShield

Auto-insurance fraud detection - model performance report
Prepared: 2026-08-09 | Dataset: insurance_fraud_data

Executive summary

This report explains how well the XGBoost model identifies claims that should be prioritised for human fraud investigation. A model score is not proof of fraud and must not be used as the sole basis for an adverse claims decision.

Performance results

Measure	Result	Plain-English meaning
Accuracy	51.02%	Overall share of correct fraud/non-fraud classifications.
Precision	30.82%	Of claims selected for review, this share was confirmed fraud in testing.
Sensitivity (recall)	79.66%	Share of known fraudulent claims detected by the model.
Specificity	41.68%	Share of genuine claims correctly left unflagged.
ROC-AUC	0.6536	Ability to rank fraudulent claims above genuine claims; 0.5 is random and 1.0 is perfect.
Review threshold	0.41	Claims at or above this probability are recommended for investigation.

Classification results

	Model: not flagged	Model: investigate
Actual genuine claim	754	1055
Actual fraudulent claim	120	470

Financial impact assessment

Measure	Value
Test claim value	AUD \$55,549,052.18
Confirmed fraud value	AUD \$13,619,959.52
Potential fraud value prioritised	AUD \$10,555,637.05

Potential value, not guaranteed savings; investigation and recovery costs are excluded.

Recommended next actions

- Prioritise high-probability claims for investigator review, while displaying claim value and supporting evidence.
- Review the threshold regularly against investigation capacity, false-positive impact, and fraud missed.
- Track investigation outcomes, recovery value, customer complaints, and model drift before retraining.
- Keep a human investigator responsible for every final claims decision and record the supporting evidence.

Governance note

Before production use, complete privacy, security, retention, access/correction, fairness, audit, and legal review. Refer to the project compliance checklist for the operational controls and escalation requirements.